

Alvaro Guerra

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EDUCATION

Texas A&M University

B.S. Mechanical Engineering

College Station, TX

2021 - 2025

EXPERIENCE

Structures Lead

IGNITORS (Rocketry Team)

11-2024 – 06-2025

College Station, TX

- Owned structural design of 11k-ft COTS rocket, performing hand calculations and ANSYS FEA to size fiberglass body tubes, couplers, and bulkheads with FOS>2
- Managed \$3.5k BOM and vendor contracts, delivering on schedule and 1% under budget.

Structures and Material Science Designer

SAE Aero Design Team

05-2022 – 04-2023

College Station, TX

- Performed weight-strength trade studies and Solidworks/ANSYS FEA for 18-ft span cargo aircraft, maximizing payload to 58% of gross weight.
- Led rapid field repair under 40°C Texas heat, sprinting materials across the flight line to ready the aircraft for a final flight that clinched 1st place overall among 40+ teams.

Garden Robot Developer

Garden Cultivation Capstone Project

08-2024 – 05-2025

College Station, TX

- Employed DFM principles to machine chassis and 3D-print custom components, finishing 100% of design goals while \$200 under budget.
- Designed and built Arduino/Pi-driven assistive garden robot; wrote embedded C++ to control drives, pumps, and sensors, boosting water efficiency 26%

Real Estate Analyst Intern

River Valley Appraisals of Texas, LLC

05-2021 – 08-2021

Edinburg, TX

- Built regression models to refine comparable sales market analysis with Excel on 100+ properties

PROJECTS

openLayup (Python)

April 2025

- Developing an open-source structural simulation tool for rocket components, leveraging Python and APIs (Onshape, SolidWorks, Granta Ansys) for material property retrieval, geometry integration, and failure analysis
- Implemented failure mode analysis using Max Stress/Strain criteria, CLT, Tsai-Wu, Tsai-Hill, Hashin-Rotem, and Puck Criterion

BeamMeUpScotty (Python, Jupyter)

April 2025

- 2-D beam/truss solver utilizing Euler-Bernoulli and Newton-Raphson; outputs GUI and interactive plots
- Implemented numerical methods for analytical estimation using Euler-Bernoulli, stiffness matrices, and Newton-Raphson techniques

TECHNICAL SKILLS

Software: SolidWorks, Onshape, ANSYS Workbench, Python (*pandas*, *NumPy*, *Matplotlib*, *BeautifulSoup*, *PyQt*), MATLAB, HTML/CSS, Git, Microsoft Office

Manufacturing: Composite lay-up & vacuum-bagging, 3D Printing (Prusa/Ender3), Hand Tools, CNC Machining, Manual Mill/Lathe

Additional: English/Spanish Fluency, TRA L1 (in progress)